In [47]:

```
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
dataset
dataset=pd.get_dummies(dataset,drop_first=True)
dataset=dataset.drop(columns=["User ID"])
dataset
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
from sklearn.preprocessing import OrdinalEncoder
encoder = OrdinalEncoder()
x_encoded = encoder.fit_transform(x).astype(int)
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x_encoded,y,test_size=1/8,random_state=42)
dataset[["Purchased"]].value_counts()
dataset.shape
print(x_train.min())
print((x_train < 0).sum())
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
0
0
```

In [48]:

```
from sklearn.naive_bayes import CategoricalNB
clf=CategoricalNB(
    alpha=1.0, force_alpha=True, fit_prior=True,
    class_prior=None, min_categories=None
)
clf.fit(x_train,y_train)
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[48]:

▾ CategoricalNB

▸Parameters

▸Fitted attributes

In [49]:

```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test, y_pred)
print("Accuracy:",accuracy * 100, "%")
```

```
[1 1 0 1 0 0 1 0 0 0 0 1 0 0 0 0 1 1 0 1 0 0 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0
 0 0 0 1 0 0 1 0 0 0 0 1 0]
Accuracy: 90.0 %
```

In [50]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[31  3]
 [ 2 14]]
```

In [51]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.94      0.91      0.93        34
           1       0.82      0.88      0.85        16

    accuracy                           0.90        50
   macro avg       0.88      0.89      0.89        50
weighted avg       0.90      0.90      0.90        50
```

In [ ]: